Assignment 1 - Git & GitHub Fundamentals

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Repository: https://github.com/douaamoutawakkildd-svg/git-session1-douaamoutawakkildd-svg

1. Repository Setup

I renamed the STARTER folder and ran `git init -b main` inside it. At first I ran it in the parent folder by mistake, so I deleted that .git folder and ran it again in the right folder. In .gitignore I put `local.env` and `*.log`, and `!important.log` so important.log is still tracked. I checked it with `git status --ignored`.

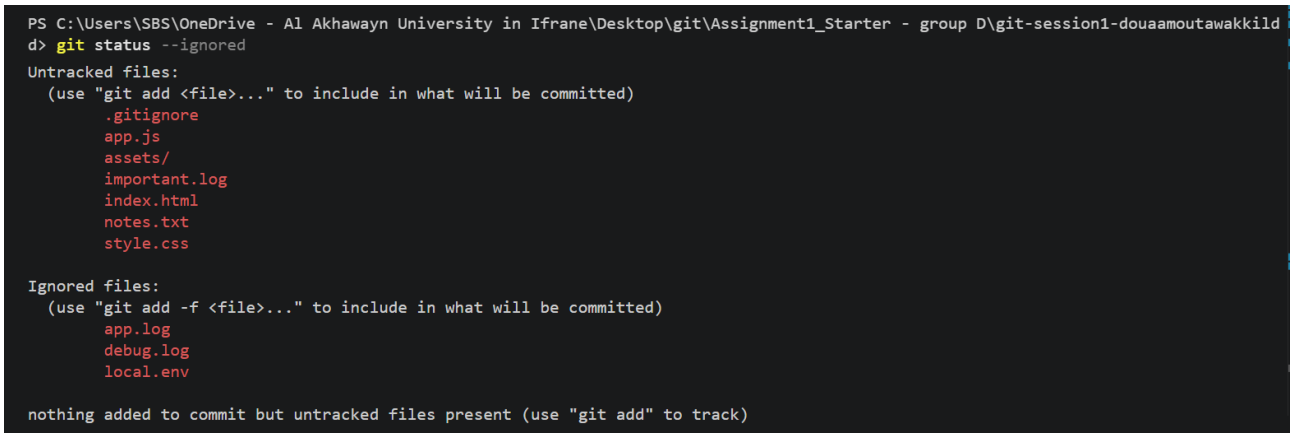


Figure 1: app.log, debug.log and local.env are ignored, important.log is still visible.

2. README & Commit History

The README has my name, GitHub username, a short description, the technologies, a code block (git clone), a link to the Git docs and the banner image from assets. I made small commits, one change each: starter files + .gitignore, README, CSS colour, app.js message, About section, merge, two README fixes (I wrote the wrong username first), the GitHub edit, and a last CSS change. The commits show the author name moutawakkildouaadd-hue because my Git email is linked to that account.

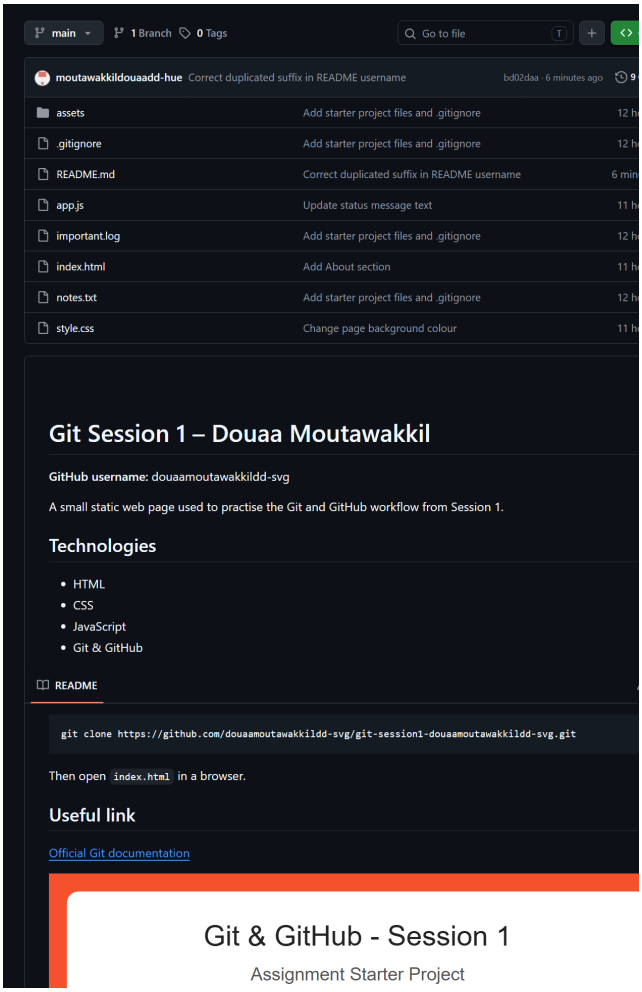


Figure 2: README rendered on GitHub.

```
d> git --no-pager log --oneline --graph
* 91a0878 (HEAD -> main) Adjust background shade
* cd798ed (origin/main, origin/HEAD) Update README.md
* bd02daa Correct duplicated suffix in README username
* a96f8b8 Fix GitHub username in README
* 08604f2 Fix GitHub username in README
*   bc8b72d Merge feature/about into main
|\
| * 1215838 (feature/about) Add About section
|/
* aa7caf0 Update status message text
* deac4f4 Change page background colour
* 58d52d3 Add README with description, technologies and banner
* ac3b547 Add starter project files and .gitignore
```

Figure 3: commit history.

3. Selective Staging

I changed style.css and app.js at the same time, then only ran `git add style.css` and committed. git status showed app.js still as modified (not staged). After that I committed app.js separately.

```
d> git status
On branch main
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   style.css

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.js
```

Figure 4: style.css staged, app.js not staged.

```
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git commit -m "Change page background colour"
[main deac4f4] Change page background colour
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.js

no changes added to commit (use "git add" and/or "git commit -a")
```

Figure 5: after the commit, app.js is still modified.

4. Branch & Merge

I created `feature/about` with `git switch -c`, added an About section in index.html and committed it on the branch. Then I went back to main and merged with `git merge --no-ff feature/about` so the merge commit shows in the graph.

```
d> git switch main
Switched to branch 'main'
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git merge --no-ff feature/about -m "Merge feature/about into main"
Merge made by the 'ort' strategy.
 index.html | 4 ++++
 1 file changed, 4 insertions(+)
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git log --oneline --graph --all
*   bc8b72d (HEAD -> main) Merge feature/about into main
|\
| * 1215838 (feature/about) Add About section
|/
* aa7caf0 Update status message text
* deac4f4 Change page background colour
* 58d52d3 Add README with description, technologies and banner
* ac3b547 Add starter project files and .gitignore
```

Figure 6: branch and merge in the log graph.

5. Remote Changes

Commit saves a snapshot in my local repository only. Push sends my local commits to GitHub. Fetch downloads the new commits from GitHub but does not change my files or my branch, it only updates origin/main. Pull is fetch + merge, so it actually brings the changes into my files. I edited the README on GitHub, ran git fetch and git status showed I was 1 commit behind, then I ran git pull.

```
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git fetch
PS C:\Users\SBS\OneDrive - Al Akhawayn University in Ifrane\Desktop\git\Assignment1_Starter - group D\git-session1-douaamoutawakkild
d> git status
On branch main
Your branch is behind 'origin/main' by 1 commit, and can be fast-forwarded.
  (use "git pull" to update your local branch)

nothing to commit, working tree clean
```

Figure 7: after fetch, main is behind origin/main by 1 commit.

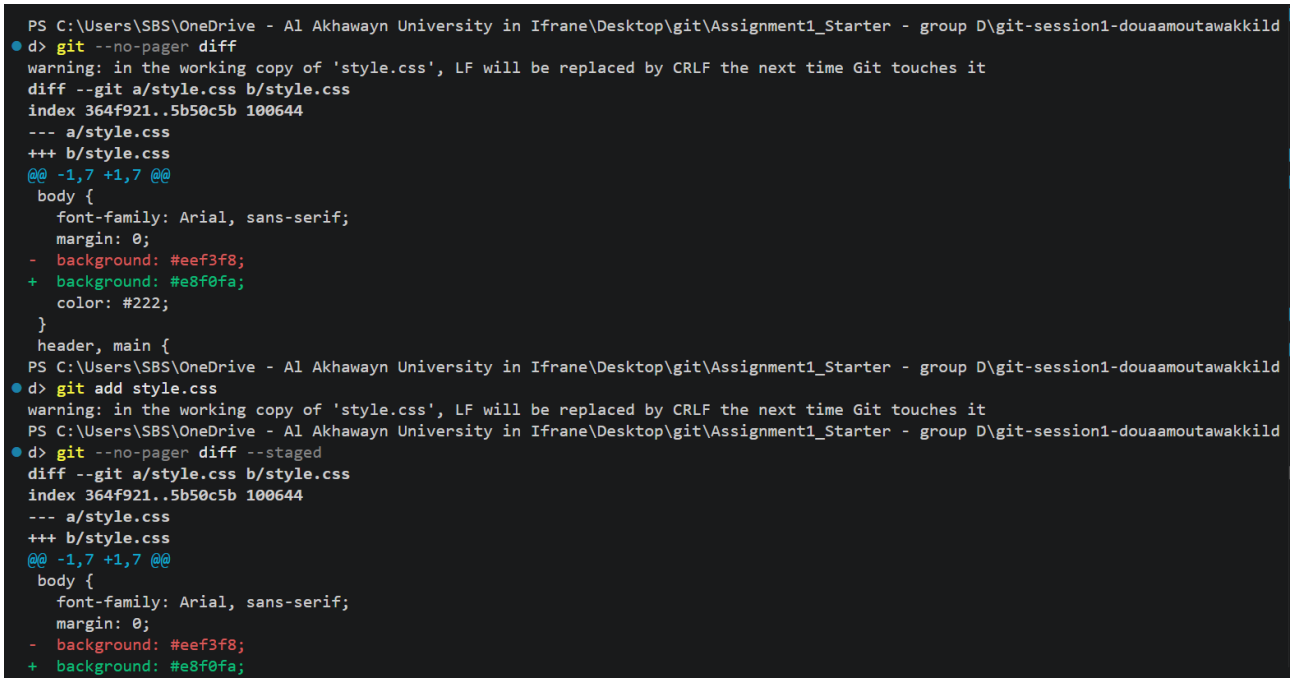


Figure 8: unstaged diff (git diff) and staged diff (git diff --staged).

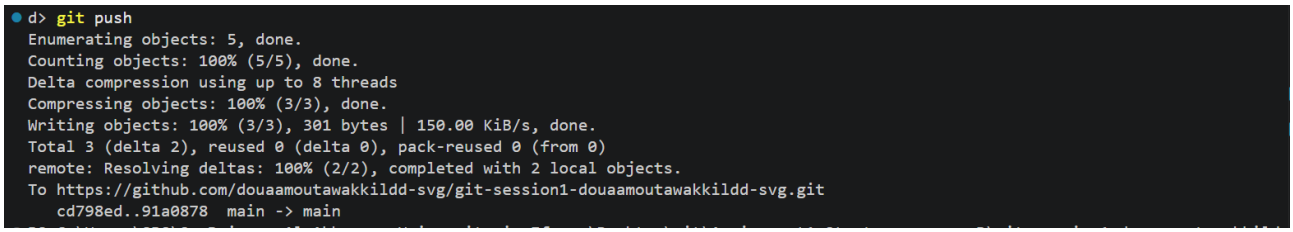


Figure 9: final push to GitHub.

6. Fair Use of AI

I used Claude (AI) as a guide while doing the assignment. I asked it which commands to use for each task and what my errors meant, for example when I ran git init in the wrong folder, when I pasted the .gitignore lines into PowerShell instead of the file, and when git diff opened in the pager. I typed and ran all the commands myself and checked each step with git status, git log and the GitHub page. Claude also helped me write this report from my screenshots; I read it and changed what was not right.

7. Sources & Reflection

Sources: Session 1 slides (Group D), Git documentation (git-scm.com/doc), GitHub Docs, Claude.

The hardest concept for me was the difference between fetch and pull, and also staged vs unstaged changes. If Git behaves unexpectedly, the first thing I would check is `git status` and which folder I am in, because most of my problems came from being in the wrong folder or a file not being saved.